

Dr Benya Wang

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CAREER PROFILE

AI & Data Platforms Engineer with a PhD and 8 years of experience designing and improving enterprise-grade data and AI platforms across mining and resources. Proven track record establishing platform standards, engineering guides, and self-service patterns that let engineering and analytics groups build independently, and partnering across data, visualisation, AI, and governance functions to turn emerging capability (GenAI, agentic systems, knowledge bases) into production-grade platform offerings. Deep expertise in MLOps, LLM / GenAI pipelines, Vision AI, and edge computing on Azure, Databricks, and AWS.

CORE COMPETENCIES

Cloud & Data Platforms

Databricks (data warehouse, workflow orchestration), Azure (AI Foundry, Azure ML, IoT Edge/Hub, Azure OpenAI, DevOps), AWS

MLOps & DevOps

MLflow, Azure Pipelines, GitHub Actions, Terraform, Bicep, Docker, CI/CD pipeline design

LLMs & Generative AI

Azure AI Foundry, Claude Code, LangChain, LightRAG, GraphRAG, Microsoft Copilot Studio, RAG pipeline design, vector stores, multi-agent orchestration, agentic platforms

ML & Data Engineering

XGBoost, LightGBM, Python, PySpark, Pandas, SQL, Power BI

Leadership & Architecture

Enterprise platform strategy, AI governance framework design, stakeholder engagement, data governance, technical mentoring

PROFESSIONAL EXPERIENCE

Specialist Machine Learning Engineer

Apr 2023 – Present

South32 · Perth, WA

- Led technical governance for the ML platform including authoring standards, engineering guides, and platform patterns; reviewing architectures; and mentoring data scientists across business units to drive consistent, production-quality delivery.
- Defined the technical strategy and built a unified AI platform (Azure AI Search, Azure OpenAI, Cosmos DB) with self-service patterns and reference implementations that reduced time-to-production for new LLM applications by ~60%, enabling multiple product teams to ship independently.
- Pioneered an air-gapped edge ML deployment pattern (Azure IoT Edge, ACR, Databricks Apps) enabling model inference on isolated industrial sites with zero internet connectivity; adopted as the enterprise-wide standard across all remote operations.
- Delivered GenAI-powered knowledge base and document intelligence capabilities (LightRAG, Azure AI Foundry) across geoscience and safety corpora, surfacing actionable insights from previously unstructured data at enterprise scale.
- Partnered with Cyber and Enterprise Architecture teams to design a Microsoft Copilot Studio governance framework and a GitHub / GitHub Copilot governance framework, establishing enterprise-wide standards for AI-assisted development, access controls, and compliant deployment; supported the organisational migration from Azure DevOps to GitHub at scale.
- Served as Acting Data Science Lead May–Jun 2026, maintaining delivery continuity across active AI platform workstreams; co-designed the platform production support model with the Operational Digital Technology team, defining ownership, escalation paths, and SLA structures.

Specialist Data Scientist

Jan 2022 – Apr 2023

South32 · Perth, WA

- Architected and led delivery of a centralised Data Science platform (Azure Databricks, MLflow, Azure Pipelines) that standardised MLOps workflows across 6+ business units, cutting the average model

deployment cycle from ~3 weeks to under 5 days and eliminating ad-hoc deployment risk.

- Led advanced feature engineering and exploratory analysis at scale using PySpark and scikit-learn across large industrial datasets.
- Partnered with data governance teams to implement stewardship and metadata management aligned with enterprise compliance policies.
- Designed and deployed a Vision AI safety system (YOLO, Supervision) for automated stop-sign monitoring, directly improving operational safety compliance.
- Implemented production multi-agent LLM pipelines (GraphRAG, AutoGen) to extract structured insights from HR records, safety incident reports, and geoscience exploration corpora.

Data Scientist

Jan 2021 – Jan 2022

Rio Tinto · Perth, WA

- Engineered serverless data pipelines (AWS Lambda, Glue, Athena) for real-time production analytics supporting operational decision-making.
- Led ML-driven optimisation of ore blending strategies, improving resource utilisation and reducing annual operational costs.
- Won an internal Hackathon for diamond image classification using TensorFlow and transfer learning.

Graduate Data Scientist

Feb 2019 – Jan 2021

Rio Tinto · Perth, WA

- Developed predictive maintenance models (LightGBM, Random Forest) for a 1,700 km rail network, reducing unplanned downtime through early fault detection.
- Containerised models in Docker and deployed via AWS SageMaker, establishing repeatable deployment and rollback processes for the team.
- Delivered spatial analytics dashboards (GIS maps, Plotly Dash) enabling operational teams to make data-driven decisions in real time.

Data Science Intern

Jan 2018 – Apr 2018

Rio Tinto · Perth, WA

- Contributed to early-stage predictive maintenance analytics and sensor data preprocessing pipelines.

EDUCATION

PhD in Environmental Engineering

Feb 2015 – Feb 2019

University of Western Australia

Thesis: Application of machine learning algorithms in urban catchments. Five peer-reviewed publications in *Geoscientific Model Development*, *Water Resources Research*, and other journals.

Bachelor of Science

Sep 2010 – Jun 2014

Zhejiang University, China

CERTIFICATIONS

Databricks Certified

Machine Learning Professional · Machine Learning Associate · Data Engineer Associate

Microsoft Certified

DevOps Engineer Expert · Azure Data Scientist Associate · Azure Data Engineer Associate · Azure Developer Associate

AWS Certified

Machine Learning – Specialty · Solutions Architect – Associate

Other

HashiCorp Certified: Terraform Associate · Anthropic: Claude Code in Action